

Module 5

Assignment- problem solving with AI (Claude + JavaScript)

Q1. JavaScript Coding Practice

- **Convert Celsius to Fahrenheit**
Write a function that takes temperature in Celsius and converts it to Fahrenheit using the formula : $F = (C / 5) * 9 + 32$.
- **Find the Factorial of a Number**
Write a function that calculates the factorial of a given non-negative integer (e.g., $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$)
- **Check the Palindrome**
Write a function that checks if a given word (e.g., "racecar") reads the same backward as forward and return true or false.
- **Sum of Array Elements**
Write a function that takes an array of numbers and returns the sum of all elements in the array.
- **FizzBuzz (Basic)**
Write a loop from 1 to 15 that prints "Fizz" for numbers divisible by 3, "Buzz" for numbers divisible by 5, "FizzBuzz" for numbers divisible by both and the number itself otherwise.

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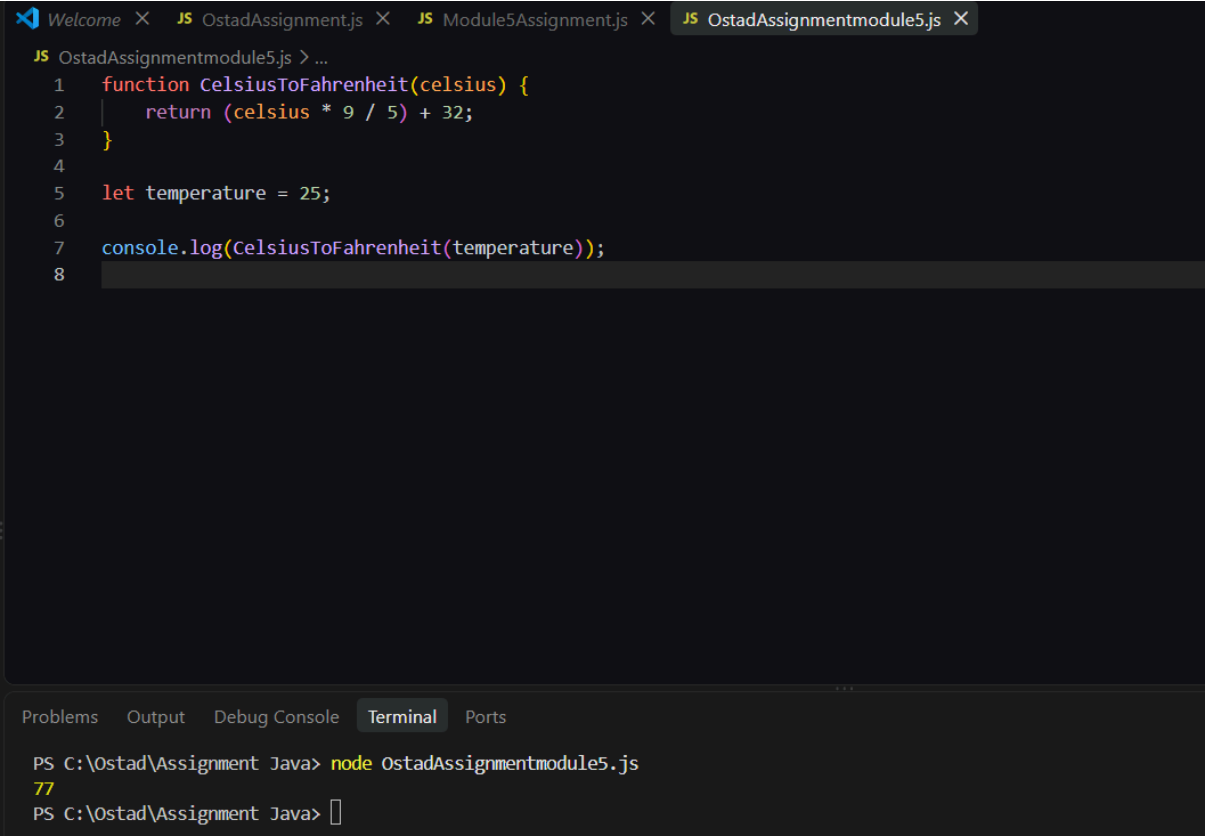
Date- 17.09.26

1. Convert Celsius to Fahrenheit

Source Code-

```
function CelsiusToFahrenheit(celsius) {  
    return (celsius * 9 / 5) + 32;  
}  
  
let temperature = 25;  
  
console.log(CelsiusToFahrenheit(temperature));
```

Input & Output-



The screenshot shows a Visual Studio Code editor with four tabs: 'Welcome', 'JS OstadAssignment.js', 'JS Module5Assignment.js', and 'JS OstadAssignmentmodule5.js'. The active tab is 'JS OstadAssignmentmodule5.js', which contains the following code:

```
JS OstadAssignmentmodule5.js > ...  
1  function CelsiusToFahrenheit(celsius) {  
2      |   return (celsius * 9 / 5) + 32;  
3      |  
4      |  
5      |  
6      |  
7      |  
8      |
```

Below the editor, the 'Terminal' panel is open, showing the command to run the file and the output:

```
PS C:\Ostad\Assignment Java> node OstadAssignmentmodule5.js  
77  
PS C:\Ostad\Assignment Java> 
```

Explanation-

This program converts a temperature from **Celsius to Fahrenheit** using the formula:

$$F = (C \times 9/5) + 32$$

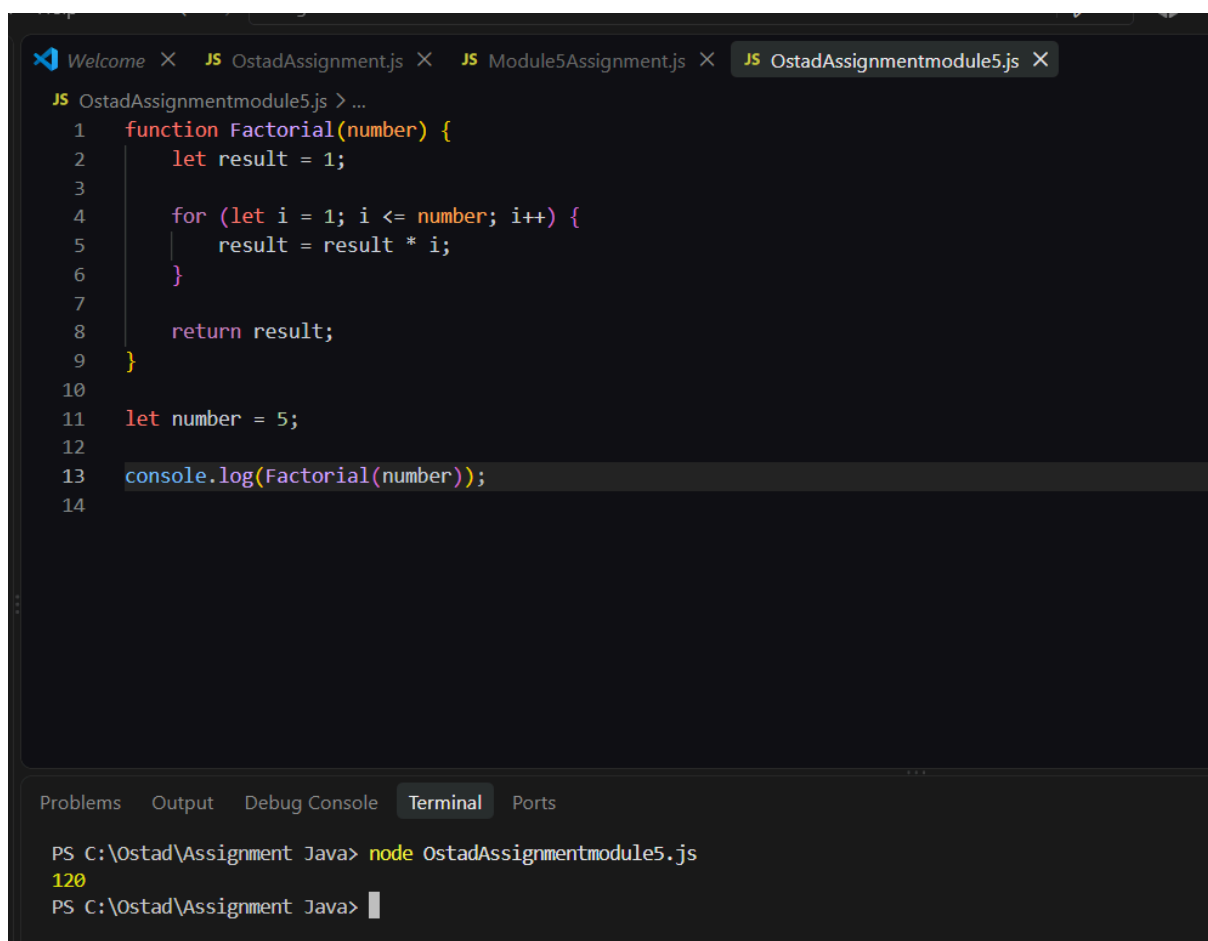
A function named `CelsiusToFahrenheit()` is created to perform the conversion. The Celsius temperature is passed to the function as a parameter. The function calculates the Fahrenheit value and returns the result. In this example, the temperature is `25°C`, which is converted to `77°F`.

2. Find the factorial number

Source Code-

```
function Factorial(number) {  
    let result = 1;  
  
    for (let i = 1; i <= number; i++) {  
        result = result * i;  
    }  
  
    return result;  
}  
  
let number = 5;  
  
console.log(Factorial(number));
```

Input & Output-



The screenshot shows a code editor with four tabs: 'Welcome', 'JS OstadAssignment.js', 'JS Module5Assignment.js', and 'JS OstadAssignmentmodule5.js'. The active tab is 'JS OstadAssignmentmodule5.js', which contains the following JavaScript code:

```
1 function Factorial(number) {  
2     let result = 1;  
3  
4     for (let i = 1; i <= number; i++) {  
5         result = result * i;  
6     }  
7  
8     return result;  
9 }  
10  
11 let number = 5;  
12  
13 console.log(Factorial(number));  
14
```

Below the code editor is a terminal window with tabs for 'Problems', 'Output', 'Debug Console', 'Terminal', and 'Ports'. The 'Terminal' tab is active, showing the command prompt output:

```
PS C:\Ostad\Assignment Java> node OstadAssignmentmodule5.js  
120  
PS C:\Ostad\Assignment Java>
```

Explanation-

This program calculates the **factorial of a non-negative integer**. A factorial is the multiplication of a number by all positive integers smaller than it.

For example:

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

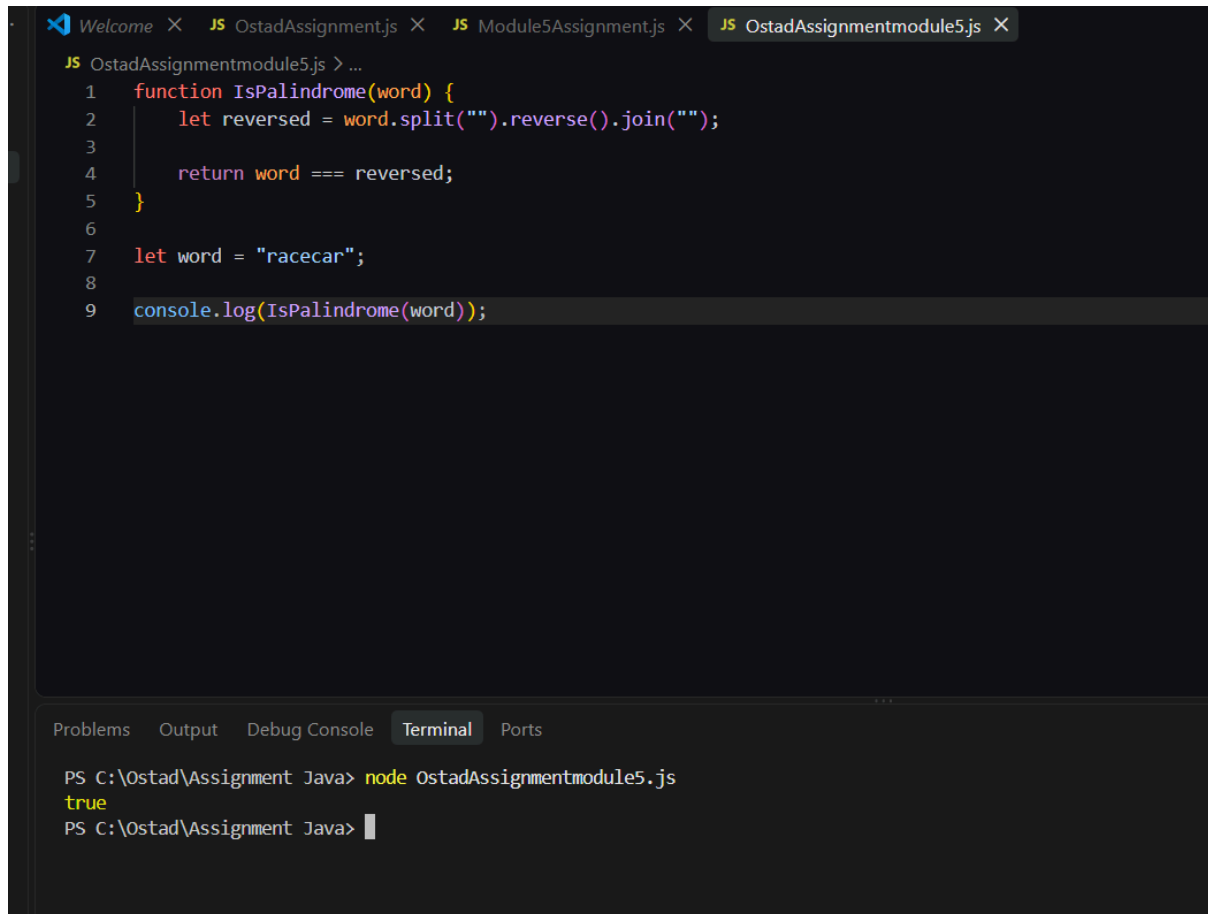
The program uses a **for** loop to multiply the numbers from **1** up to the given number. The variable **result** stores the multiplication result. Finally, the function returns the factorial.

3. Check for Palindrome

Source Code-

```
function IsPalindrome(word) {  
    let reversed = word.split("").reverse().join("");  
  
    return word === reversed;  
}  
  
let word = "racecar";  
  
console.log(IsPalindrome(word));
```

Input & Output-



```
JS OstadAssignmentmodule5.js > ...
1  function IsPalindrome(word) {
2      let reversed = word.split("").reverse().join("");
3
4      return word === reversed;
5  }
6
7  let word = "racecar";
8
9  console.log(IsPalindrome(word));
```

```
Problems  Output  Debug Console  Terminal  Ports
PS C:\Ostad\Assignment Java> node OstadAssignmentmodule5.js
true
PS C:\Ostad\Assignment Java> 
```

Explanation-

This program checks whether a given word is a palindrome. A palindrome is a word that reads the same forward and backward.

For example:

racecar → racecar

The program reverses the given word using `split()`, `reverse()`, and `join()`. It then compares the original word with the reversed word using the strict equality operator `===`.

If both are the same, the function returns `true`; otherwise, it returns `false`.

4. Sum of Array Elements

Source Code-

```
function SumArray(numbers) {  
    let sum = 0;  
  
    for (let number of numbers) {  
        sum = sum + number;  
    }  
  
    return sum;  
}  
  
let numbers = [10, 20, 30, 40, 50];  
  
console.log(SumArray(numbers));
```

Input & Output-

```
JS OstadAssignmentmodule5.js > ...
1  function SumArray(numbers) {
2      let sum = 0;
3
4      for (let number of numbers) {
5          sum = sum + number;
6      }
7
8      return sum;
9  }
10
11  let numbers = [10, 20, 30, 40, 50];
12
13  console.log(SumArray(numbers));
```

Problems Output Debug Console **Terminal** Ports

PS C:\Ostad\Assignment Java> node OstadAssignmentmodule5.js
150
PS C:\Ostad\Assignment Java>

Explanation-

This program calculates the **sum of all numbers in an array**.

The function `SumArray()` receives an array of numbers as a parameter. A `for...of` loop is used to go through each element of the array. Each number is added to the `sum` variable. After all elements have been processed, the function returns the total.

For the given array:

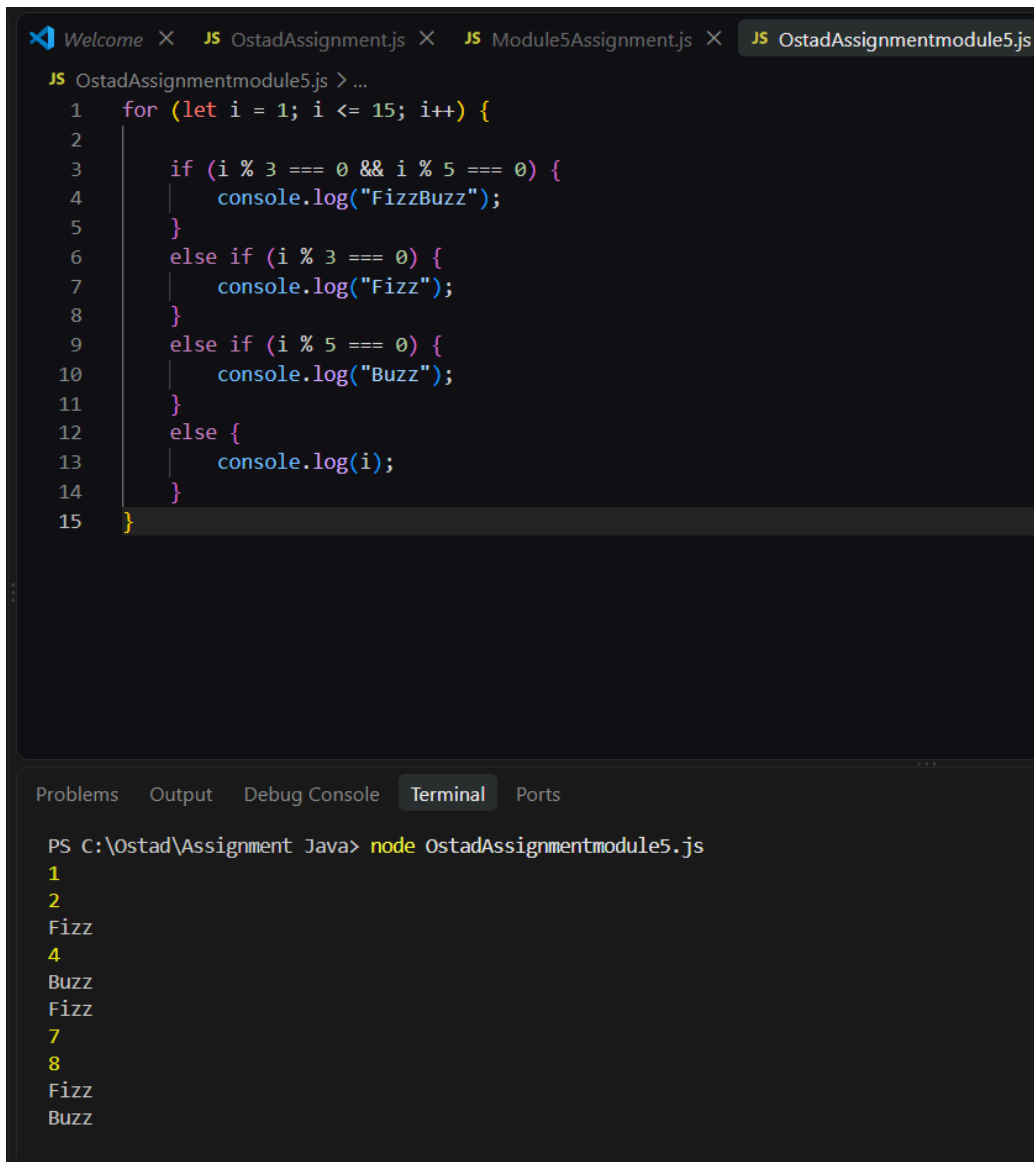
$$10 + 20 + 30 + 40 + 50 = 150$$

5. FizzBazz (Basic)

Source Code-

```
for (let i = 1; i <= 15; i++) {  
  
    if (i % 3 === 0 && i % 5 === 0) {  
        console.log("FizzBuzz");  
    }  
    else if (i % 3 === 0) {  
        console.log("Fizz");  
    }  
    else if (i % 5 === 0) {  
        console.log("Buzz");  
    }  
    else {  
        console.log(i);  
    }  
}
```

Input & Output-

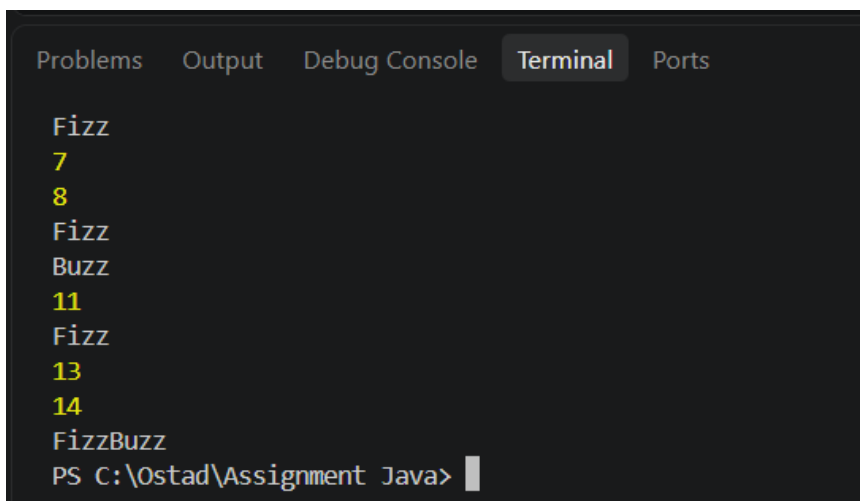


The screenshot shows the VS Code editor with three tabs: 'Welcome', 'JS OstadAssignment.js', and 'JS Module5Assignment.js'. The active tab is 'JS OstadAssignmentmodule5.js', which contains the following JavaScript code:

```
JS OstadAssignmentmodule5.js > ...
1  for (let i = 1; i <= 15; i++) {
2
3      if (i % 3 === 0 && i % 5 === 0) {
4          console.log("FizzBuzz");
5      }
6      else if (i % 3 === 0) {
7          console.log("Fizz");
8      }
9      else if (i % 5 === 0) {
10         console.log("Buzz");
11     }
12     else {
13         console.log(i);
14     }
15 }
```

Below the editor, the 'Terminal' tab is active, showing the command to run the file and its output:

```
PS C:\Ostad\Assignment Java> node OstadAssignmentmodule5.js
1
2
Fizz
4
Buzz
Fizz
7
8
Fizz
Buzz
```



This is a close-up of the terminal window from the previous screenshot, showing the output of the FizzBuzz program for numbers 1 through 15. The output is as follows:

```

Fizz
7
8
Fizz
Buzz
11
Fizz
13
14
FizzBuzz
PS C:\Ostad\Assignment Java>
```

Explanation-

This program uses a `for` loop to check the numbers from **1 to 15**.

The modulus operator `%` is used to check whether a number is divisible by 3 or 5.

- Numbers divisible by **both 3 and 5** print "`FizzBuzz`".
- Numbers divisible by **3** print "`Fizz`".
- Numbers divisible by **5** print "`Buzz`".
- All other numbers print the number itself.

The condition for both 3 and 5 is checked first so that **15** is correctly displayed as "`FizzBuzz`".